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question 1

a and b writing 4 records into the csv file

In [17]:
```
import csv
import os
data=[['Student_name','cgpa','Certification_count','internship_exp_yrs'],
     ['kishori',9.5,100,3],['rajavel',9.2,100,5],['nideesh',9.0,50,1],['bharani',9.1,75,2]]#defing the data for the
with open("student.csv","w")as file:
    writer=csv.writer(file)
    writer.writerows(data)#writing multiple rows in the "student.csv"
```

checking whether the data is written on the student.csv

In [18]:
```
with open("student.csv","r")as file:#opening the file in read mode
    students=csv.reader(file)#storing the content in thes tudents variable
    for student in students:
      print(student)#printing the data
```

```
['Student_name', 'cgpa', 'Certification_count', 'internship_exp_yrs']
['kishori', '9.5', '100', '3']
['rajavel', '9.2', '100', '5']
['nideesh', '9.0', '50', '1']
['bharani', '9.1', '75', '2']
```

c.printing the student.csv path

In [19]:
```
path=os.path.abspath("student.csv")
print(path)
```

```
/home/matlab/student.csv
```

d.writing another 4 records one by one

In [54]:
```
with open ("student.csv","w")as file:
    writer=csv.writer(file)
    writer.writerow(['Student_name','cgpa','Certification_count','internship_exp_yrs'])
    writer.writerow(['sharmila_mam',10.0,500,10])
    writer.writerow(['viswanatham_sir',9.5,150,40])
    writer.writerow(['saraswathi_mam',9.7,20,7])
    writer.writerow(['kamali',9.7,90,3.5])
```

checking whether the another 4 records is stored or not

In [21]:
```
with open("student.csv","r")as file:#opening the file in read mode
    students=csv.reader(file)#storing the content in thes tudents variable
    for student in students:
      print(student)#printing the data
```

```
['Student_name', 'cgpa', 'Certification_count', 'internship_exp_yrs']
['sharmila_mam', '10.0', '500', '10']
['viswanatham_sir', '9.5', '150', '40']
['saraswathi_mam', '9.7', '20', '7']
['kamali', '9.7', '90', '3.5']
```

becasuse of opening the file as in writer mode if you try to write another set of data the new data will overwrite the old data

e. Write the same data as Key Value Pairs using DictWriter and

save in students_dict.cs

In [46]:
```
dict_data=[{'student_name':'student_name','cgpa':'cgpa','certification_count':'certification_count','internship_yrs
    {'student_name': 'sharmila_mam','cgpa':10,'certification_count':500,'internship_yrs':10},
          {'student_name': 'viswanatham_sir','cgpa':9,'certification_count':150,'internship_yrs':40},
          {'student_name': 'saraswathi_mam','cgpa':9,'certification_count':150,'internship_yrs':7},
{'student_name': 'kamali','cgpa':9,'certification_count':90,'internship_yrs':3.5}]

fieldname={'student_name','cgpa','certification_count','internship_yrs'}

with open ("student_dict.csv","w")as file:
    writer=csv.DictWriter(file,fieldnames=fieldname)
    writer.writerows(dict_data)
```

checking whether the data is written or not

In [47]:
```
with open ("student_dict.csv","r")as file:
    reader=csv.DictReader(file)
    for student in reader:
        print(student)
```

```
{'internship_yrs': '10', 'certification_count': '500', 'student_name': 'sharmila_mam', 'cgpa': '10'}
{'internship_yrs': '40', 'certification_count': '150', 'student_name': 'viswanatham_sir', 'cgpa': '9'}
{'internship_yrs': '7', 'certification_count': '150', 'student_name': 'saraswathi_mam', 'cgpa': '9'}
{'internship_yrs': '3.5', 'certification_count': '90', 'student_name': 'kamali', 'cgpa': '9'}
```

g. Change the mode from w to a and append three more records to the first file

In [56]:
```
data=[['kishori',9.5,100,3],['rajavel',9.2,100,5],['nideesh',9.0,50,1]]#defing the data for the file and i am stori
with open("student.csv","a")as file:
    writer=csv.writer(file)
    writer.writerows(data)#appending  multiple rows in the "student.csv"
```

h. Read the rows from the first csv file and print them all

In [57]:
```
with open("student.csv","r")as file:#opening the file in read mode
    students=csv.reader(file)#storing the content in thes tudents variable
    for student in students:
      print(student)#printing the data
```

```
['Student_name', 'cgpa', 'Certification_count', 'internship_exp_yrs']
['sharmila_mam', '10.0', '500', '10']
['viswanatham_sir', '9.5', '150', '40']
['saraswathi_mam', '9.7', '20', '7']
['kamali', '9.7', '90', '3.5']
['kishori', '9.5', '100', '3']
['rajavel', '9.2', '100', '5']
['nideesh', '9.0', '50', '1']
```

QUESTION 2

In [62]:
```
import pandas as pd
df=pd.read_csv("pandas_file.csv")
print(df)
```

```
  Student_name  cgpa  Certification_count  internship_exp_yrs
0      kishori   9.5                  100                   3
1      rajavel   9.2                  100                   5
2      nideesh   9.0                   50                   1
3      bharani   9.1                   75                   2
```

QUESTION 3

In [74]:
```
import json
dict_data={
    {'student_name': 'sharmila_mam','cgpa':10,'certification_count':500,'internship_yrs':10},
          {'student_name': 'viswanatham_sir','cgpa':9,'certification_count':150,'internship_yrs':40},
          {'student_name': 'saraswathi_mam','cgpa':9,'certification_count':150,'internship_yrs':7},
```

```
{'student_name': 'kamali','cgpa':9,'certification_count':90,'internship_yrs':3.5}}

fieldname={'student_name','cgpa','certification_count','internship_yrs'}

with open ("student_dict.json","w")as file:
    writer=json.dump(file,dict_data)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[74], line 2
      1 import json
----> 2 dict_data={
      3     {'student_name': 'sharmila_mam','cgpa':10,'certification_count':500,'internship_yrs':10},
      4           {'student_name': 'viswanatham_sir','cgpa':9,'certification_count':150,'internship_yrs':40},
      5           {'student_name': 'saraswathi_mam','cgpa':9,'certification_count':150,'internship_yrs':7},
      6 {'student_name': 'kamali','cgpa':9,'certification_count':90,'internship_yrs':3.5}}
      8 fieldname={'student_name','cgpa','certification_count','internship_yrs'}
     10 with open ("student_dict.json","w")as file:

TypeError: unhashable type: 'dict'
```

c. Iterate through the JSON file and print the contents

In [86]:
```
with open("student.json","r")as file:#opening the file in read mode
    students=json.load(file)#storing the content in thes tudents variable
    print(students)
```

```
{'student_name': 'sharmila_mam', 'cgpa': 10, 'certification_count': 500, 'internship_yrs': 10}
```

In [ ]:
```
with open("student.json","r")as file:#opening the file in read mode
    students=json.load(file)#storing the content in thes tudents variable
    print(students['student_name'])
```

In [84]:
```
st={'student_name': 'sharmila_mam','cgpa':10,'certification_count':500,'internship_yrs':10}
writer=json.dumps(st)
print(st['student_name'])
```

```
sharmila_mam
```

In [ ]: